

INNOVATION MANAGEMENT - PROJECT

LECTURER: Dr. Velimir Srica

STUDENT: Balaji Kendre

SMARTPHONE SECURITY DESIGN THINKING SOLUTION

ENTERPRISE BUSINESS INNOVATION PROJECT

THE PROBLEM



- People depend on realistic security available on smartphones and tablets, but these built-in features may not be sufficient to protect against common threats such as theft and loss, hacking, cybersecurity attacks, risky downloadable applications, unsafe networks and Wi-Fi, eavesdropping attacks, spam and unknown-number messaging, and the need for backup and restoration of data, as well as data security and privacy.



- Using a global estimate- about 251 million smartphones are lost or stolen each year.
- Global cited estimate- Over 1.7 billion people are victimized each year by cyberattacks, online fraud, and related online scams.

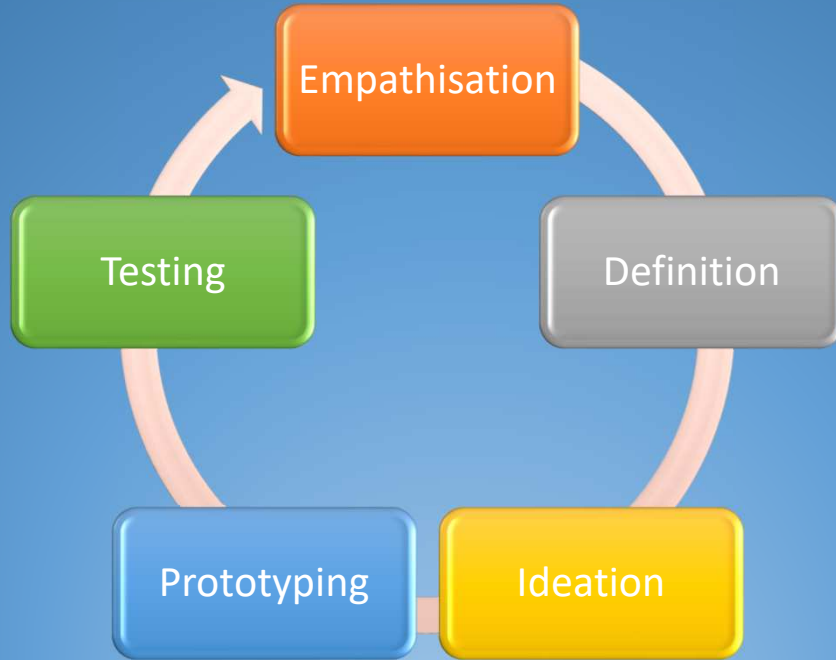


- About 2.4 billion children and young people under 18 have access to smartphones and online activities, though the unsecure level of supervision, safety controls, and access restrictions only varies by culture, family and device settings.

SmartSupervision AI



DESIGN THINKING PROCESS



Empathisation



How can we help people to protect their security.



Definition

Current solutions

- Purchase insurance to help cover loss, theft, or accidental damage.
- Install security software (reputable antivirus/mobile protection).
- Use strong security settings, such as a screen lock and secure app permissions.
- Avoid unknown contacts—do not answer or respond to suspicious calls, spam, or messages.
- Stay alert while using your phone and watch for unusual behavior.
- Back up your device regularly or enable automatic backups if available.
- Browse safely and responsibly, downloading apps only when you're confident they are legitimate.
- Be cautious with links and websites—avoid untrusted sites and never click unknown links.

These insights inspired the development of SmartSupervision AI
for smarter protection of your smartphone.

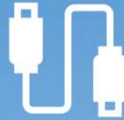
Ideation

How to solve the problem?

- ✓ Develop an AI-powered switching data centre where live traffic is monitored in real time by AI-power agents.
- ✓ Create a SmartSupervision AI app that monitors and manages endpoint device activity to help protect endpoint smartphone security.
- ✓ Integrate an AI-powered switching data centre specifically designed for detection, prevention and real time monitoring.

Features integration at AI switching centre

- AI-assisted standards and frameworks:-
Develop standards for AI-assisted smart device management, including policies, roles, reporting, and automation workflows.
- Live traffic correlation and user-activity analysis:-
Correlate network and telemetry signals in near-real time to analyze user/device behaviour and detect anomalies.
- Real-time traffic monitoring with intrusion detection/prevention:-
Implement continuous traffic monitoring with built-in IDS/IPS capabilities and alerting/response automation.
- AI-driven security controls:-
Use AI to prevent unauthorized access, reduce the risk of data breaches, and support automated containment actions

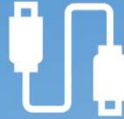


Features integration at SmartSupervision AI App

- SmartApp activity management:-
One app to monitor and manage activity across all supported apps and endpoints.
- VPN integration:-
Built-in VPN support for secure connectivity and policy enforcement.
- Exact live location with map view:-
Real-time location tracking integrated with Google Maps for guided supervision and incident response.
- Background operation:-
App continues running in the background to support ongoing monitoring and timely notifications.



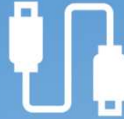
- Patch management and vulnerability mitigation:-
Continuously assess device/software exposure and prioritize patching against known vulnerabilities and active exploitation.
- Protection against mobile attack vectors:-
Detect and mitigate threats affecting mobile endpoints, including suspicious apps, tampering, privilege escalation, and compromised device behaviour.
- Data backup and restoration:-
Provide automated backup, recovery testing, and restoration procedures to minimize downtime and data loss.
- Mobile Device Management (MDM):-
Centralize device enrollment, configuration, compliance enforcement, and lifecycle management.



- AI voice integration:-
Voice interactions for end users (e.g., alerts, guidance, and simple commands), with role-appropriate permissions.
- MDM services for corporate groups:-
Mobile Device Management capabilities for corporate/group users (enrollment, configuration, compliance, and policy control).
- Security features integration:-
Integrate practical protections such as secure communications, threat detection, and account safeguards.
- Subscriber authentication:-
User login and verification to ensure only valid subscribers can access services.



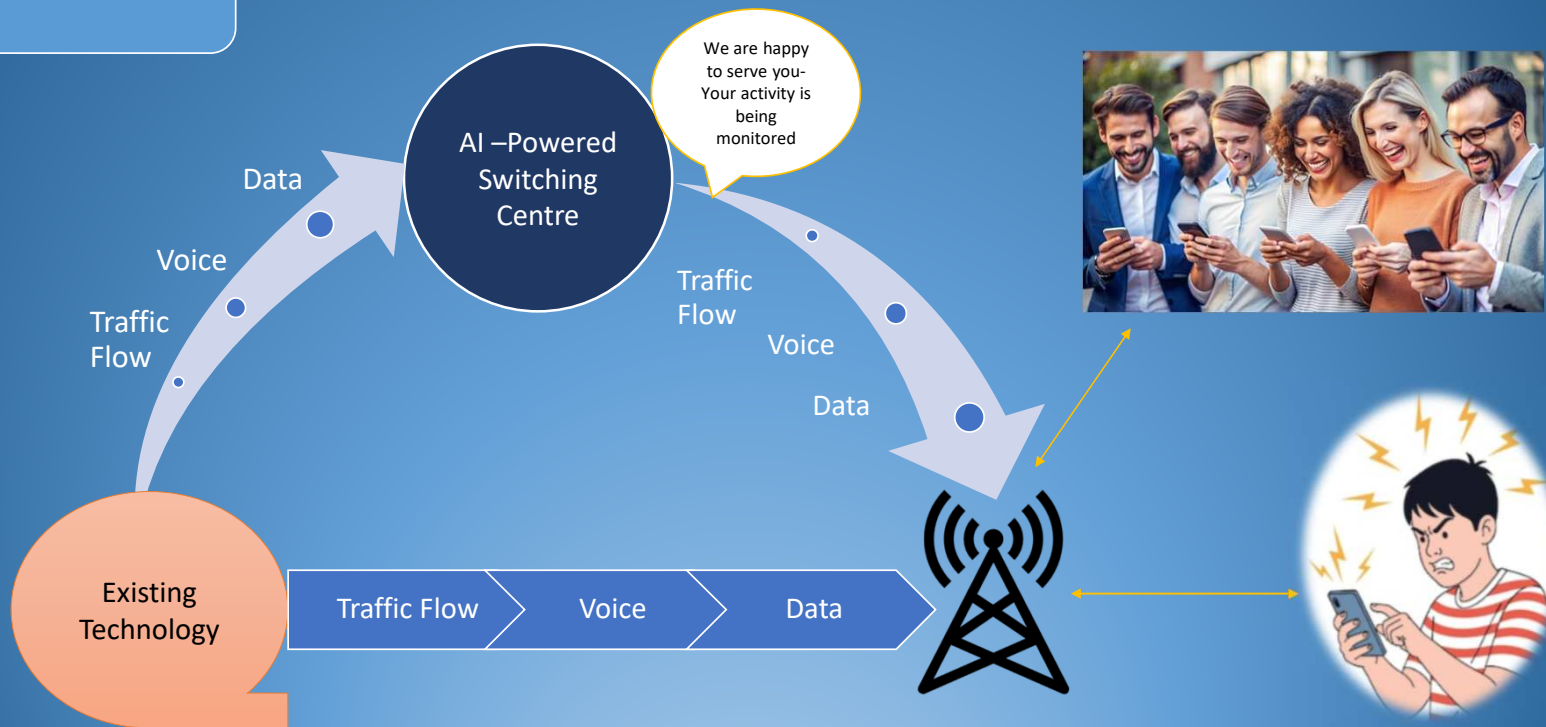
- AI voice-assistance for endpoint operations:-
Define voice-guided guidelines and workflows for monitoring, incident triage, and managing endpoint activity (with role-based controls and audit logs).



- Administrative features for policy holders:-
Admin console features for insurance policy holders (manage coverage settings, view reports/alerts, and control related supervision options).

Prototyping

Artificial Intelligence Implementation (Prototype + Testing)





Testing

Positive comments:

- Strong value focus on security outcomes.
- Good systems thinking.
- Coverage of common mobile threats.
- Centralized supervision concept is appealing.
- Include resilience features.
- Insurance- administration angle adds business relevance.
- Prototype potential is high.

Constructive criticism:

- Be specific about scope.
- Keep the architecture secure by design.
- Address false positives and add strategy.
- Plan for platform constraints. Mobile OS restrictions can prevent deep inspection.
- Reduce single point of failure concern.
- Write down compliance and retention assumptions.
- Measure outcomes, not just features.

Questions that have arisen:

- What exactly SmartSupervision-AI App see and control on mobile OS?
- How do you define a threat and what triggers prevention?
- How do you prevent false positives from causing disruption?
- How will system behave during datacentre latency or outages?
- How will MDM policies be secured end to end encryption?
- How will backup/restore work securely after compromise?
- How will you support different device types and OS versions.

Suggestions:

- Strengthen trust boundaries.
- Use a staged response model instead of blocking.
- Implement allowlists and safe-mode for critical apps.
- Total control describe the achievable control.
- Make the AI practical with prioritization engine.
- Design for high availability of enforcement.
- Measure success metrics in your proposal.

Conclusion



Improves AI-integrated security governance across mobile devices and user activity.



Reduces vulnerability exposure through smarter, continuous monitoring.



Strengthens protection against modern mobile threats.



Provides AI-driven guidance that helps individuals manage security settings and respond safely to risks.



Overall, the project concludes that AI-driven monitoring with automated security actions improves device governance, threat prevention, and operational continuity.



Delivers a more secure smartphone experience than existing technologies by enabling real-time detection, faster prevention, and integrated endpoint controls.

Future addition

IoT Devices

- Expand AI smart monitoring to IoT devices with real-time threat detection:
Expand the AI smart-monitoring capability to IoT network devices, enabling automated detection of unsafe behaviour and suspicious device connections in real time.

Smart Devices

- Create multi-device standards/frameworks for smartphones, tablets, PCs, and laptops:
Develop standards and frameworks for multi-device scalability, so the platform can support smartphones, tablets, PCs, and laptops with consistent security policies and management controls.

Live CCTV

- Add optional live CCTV monitoring integrated with security alerts:
Extend SmartSupervision with an optional live CCTV monitoring feature for individuals and organizations, integrated with threat detection and security alert workflows.

